

XF vs CF in AEM

jcr:primaryType · JCR Structure · Key Differences

Experience Fragment vs Content Fragment — Complete JCR Comparison

This guide explains the JCR node structure of Experience Fragments and Content Fragments, focusing on their **jcr:primaryType** values and why they differ fundamentally.

1. The Direct Answer — jcr:primaryType Values

Experience Fragment — jcr:primaryType

An Experience Fragment is treated as a **Page** in AEM. So it uses the same primary types as any AEM page.

```
/content/experience-fragments/mysite/my-xf
  ■■■ jcr:primaryType = cq:Page          ← root node is a PAGE
  ■■■ jcr:content
    ■■■ jcr:primaryType = cq:PageContent ← content node
    ■■■ sling:resourceType = "cq/experience-fragments/editor/..."
    ■■■ ... components, parsys, layout inside
```

jcr:primaryType at root → cq:Page jcr:primaryType at jcr:content → cq:PageContent

Content Fragment — jcr:primaryType

A Content Fragment is treated as a **DAM Asset**. It lives in the DAM just like an image or video, but contains structured text fields.

```
/content/dam/mysite/articles/my-article
  ■■■ jcr:primaryType = dam:Asset          ← root node is a DAM ASSET
  ■■■ jcr:content
    ■■■ jcr:primaryType = dam:AssetContent ← asset content node
    ■■■ contentFragment = true             ← flag marking it as a CF
    ■■■ data
      ■■■ jcr:primaryType = nt:unstructured
```

jcr:primaryType at root → dam:Asset jcr:primaryType at jcr:content → dam:AssetContent

2. Full JCR Structure Comparison

Experience Fragment — Full Structure

```
/content/experience-fragments/mysite/hero-banner
  ■■■ jcr:primaryType      = cq:Page
  ■■■ jcr:createdBy        = "admin"
  ■■■ jcr:content
    ■■■ jcr:primaryType    = cq:PageContent
    ■■■ jcr:title          = "Hero Banner XF"
    ■■■ sling:resourceType = "cq/experience-fragments/editor/components/..."
    ■■■ cq:template        = "/libs/cq/experience-fragments/templates/..."
    ■■■ cq:lastModified    = "2024-06-16T10:00:00"
    ■■■ root (parsys — holds actual components)
      ■■■ image_component
        ■■■ fileReference = "/content/dam/image.jpg"
      ■■■ text_component
        ■■■ text = "<p>Welcome!</p>"
```

Content Fragment — Full Structure

```
/content/dam/mysite/articles/my-article
  ■■■ jcr:primaryType = dam:Asset
  ■■■ jcr:content
    ■■■ jcr:primaryType = dam:AssetContent
    ■■■ contentFragment = true
    ■■■ cq:lastModified = "2024-06-16T10:00:00"
    ■■■ data
      ■■■ jcr:primaryType = nt:unstructured
      ■■■ cq:model → "/conf/mysite/settings/dam/cfm/models/article"
      ■■■ master (default variation)
        ■■■ title = "My Article Title"
        ■■■ body = "Full article body text..."
        ■■■ author = "/content/dam/authors/john-doe"
```

3. Side-by-Side Comparison

Property	Experience Fragment	Content Fragment
jcr:primaryType (root)	cq:Page	dam:Asset
jcr:primaryType (jcr:content)	cq:PageContent	dam:AssetContent
Stored under	/content/experience-fragments/	/content/dam/
Has components / layout?	YES (parsys, grid)	NO — pure data fields
Has presentation / HTML?	YES — renders HTML	NO — channel agnostic

Property	Experience Fragment	Content Fragment
Has variations?	YES (web, email, etc.)	YES (master, custom)
Identified by	slings:resourceType	contentFragment = true
Managed in	Sites console	Assets / DAM console
Delivered as	Rendered HTML	JSON via GraphQL / API
Use case	Reusable designed UI blocks	Reusable raw content

4. Why Are They Different?

Experience Fragment = a PAGE with layout and components

It needs to render HTML, hold components like images and text, support page templates and parsys — so AEM treats it exactly like a page. Hence **cq:Page** and **cq:PageContent**.

Content Fragment = a DAM ASSET with structured fields

It only stores raw structured data (title, body, author, date). It has no idea how to render itself — that's the frontend's job. AEM stores it in the DAM alongside images and documents. Hence **dam:Asset** and **dam:AssetContent**.

```

Experience Fragment  → "Here is a designed Hero Banner"
                      Has HTML, CSS, components – ready to render
                      jcr:primaryType = cq:Page / cq:PageContent

Content Fragment     → "Here is an Article with title, body, author"
                      Has raw data only – no rendering logic
                      jcr:primaryType = dam:Asset / dam:AssetContent

```

5. How to Identify Each in Code

Check if a resource is a Content Fragment

```

Resource resource = resourceResolver.getResource("/content/dam/mysite/articles/my-article");

// Method 1: adaptTo ContentFragment API
ContentFragment cf = resource.adaptTo(ContentFragment.class);
if (cf != null) {
    // it IS a Content Fragment
    String title = cf.getElement("title").getContent();
}

// Method 2: check the contentFragment flag
Resource jcrContent = resource.getChild("jcr:content");
ValueMap props = jcrContent.getValueMap();
boolean isCF = props.get("contentFragment", false);

```

```
// → true for Content Fragments, false for regular assets
```

Check if a resource is an Experience Fragment

```
Resource resource = resourceResolver.getResource(
    "/content/experience-fragments/mysite/hero-banner"
);

// Check jcr:primaryType
String primaryType = resource.getValueMap()
    .get("jcr:primaryType", String.class);
// → "cq:Page" for Experience Fragments

// Or adapt to Page
Page xfPage = resource.adaptTo(Page.class);
if (xfPage != null) {
    // it IS a page (could be XF or regular page)
    String template = xfPage.getProperties()
        .get("cq:template", String.class);
    // XF template path contains "experience-fragments"
}
```

6. Quick Reference — jcr:primaryType Cheat Sheet

Node Type	jcr:primaryType	Where
AEM Page	cq:Page	/content/mysite/...
Page content node	cq:PageContent	/content/mysite/.../jcr:content
Experience Fragment	cq:Page	/content/experience-fragments/...
XF content node	cq:PageContent	/content/experience-fragments/.../jcr:content
Content Fragment	dam:Asset	/content/dam/...
CF content node	dam:AssetContent	/content/dam/.../jcr:content
DAM Image/Asset	dam:Asset	/content/dam/...
Generic node	nt:unstructured	anywhere (data, config nodes)
CF Model	nt:unstructured	/conf/.../dam/cfm/models/...